

Mitchell Munns | mitchmunns@ucsb.edu | (630)809-0142 | Psychological & Brain Sciences |
University of California, Santa Barbara | Santa Barbara, CA, 93106

Education

PhD, Psychology , Cognition, Perception, and Cognitive Neuroscience <i>University of California, Santa Barbara</i> , Advisor: Dr. Mary Hegarty	2022 – 2027 (expected)
Bachelor of Science, Psychology <i>Northern Illinois University</i> , Departmental Honors, Summa Cum Laude	2009 – 2010; 2019 – 2021
Associate of Science, Mathematics <i>Waubonsee Community College</i>	2007 – 2009

Research

Graduate Student Researcher <i>University of California, Santa Barbara</i> ,	2022 – 2025
Lab Manager (Junior Specialist) <i>University of California, Santa Barbara</i> ,	2021 – 2022
Undergraduate Research Assistant <i>Northern Illinois University</i>	2019 – 2021
Student Coordinator <i>CREATE Center, Northern Illinois University</i>	2019 – 2020

Teaching

Teaching Assistant <i>University of California, Santa Barbara</i>	2022 – present
Statistical Methods in Psychological & Brain Sciences	2026 Winter
Research Methods Psychological & Brain Sciences	2025 Fall
Laboratory in Memory and Cognition	2024 Fall
Applications of Cognitive Psychology	2022 Fall

Awards & Honors

Editor's Choice Award (<i>Journal of Intelligence</i>)	2023
Conference Travel Grant (UCSB)	2023, 2025
Outstanding Research/Scholar Award (NIU)	2021
Woodruff Scholarship (NIU)	2019 – 2021
Best Presentation (NIU Psychology undergraduate conference)	2021
Presidential Scholarship (NIU)	2009 – 2010
WYSE Team competition scholarship (Waubonsee Community College)	2007 – 2009

Publications

Hegarty, M., Topete, A., Zhou M., Munns, M. E. , Mayer, B. (in press) Learning Spatial Layout in Virtual Environments. In Plass, J., Mayer, R. & Makransky, G. (Eds.), <i>The Handbook of Learning in Virtual Reality</i> . MIT Press.	
*Martella, A. M., *Lawson, A. P., Martella, R. C., LaBonte, K., Zhao, F., Delgado, C. Y., Munns, M. E. , Wells LeRoy, A., Gluck, J. A. & Mayer, R. E. (2026). Systematically Reviewing the Rigour of Immersive Virtual Reality Research in STEM Education: A Deep Dive Into Threats to Internal Validity. <i>Journal of Computer Assisted Learning</i> , 42(1), e70165.	DOI
*Lawson, A. P., *Martella, A. M., LaBonte, K., Delgado, C. Y., Zhao, F., Gluck, J. A., Munns, M. E. , Wells LeRoy, A., Mayer, R. E. (2024). Confounded or Controlled? A Systematic Review of Media Comparison Studies in Immersive Virtual Reality Research in STEM Education. <i>Educational Psychology Review</i> , 36(3), 1-35.	DOI
*Varshney, A., * Munns, M. E. , Kasowski, J., Zhou, M., He, C., Grafton, S. T., Giesbrecht, B., Hegarty, M., & Beyeler, M. (2024). Stress affects navigation strategies in immersive virtual reality. <i>Scientific Reports</i> , 14(1), 5949.	DOI *authors contributed equally to this work

Munns, M. E., Kunz, A., He, C., & Hegarty, M. (2023). Measures of Visualizing Cross Sections of 3D Objects: An Assessment Using Item Response Theory. *Journal of Intelligence*, 11(11), 205. [DOI](#)

Munns, M. E., Tranquada-Torres, B., Chrastil, E., & Hegarty, M. (2022). Large-Scale vs Small-Scale Spatial Abilities: Development of a Broad Spatial Activities Questionnaire. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 44. [DOI](#)

Submitted

Munns, M. E., & Hegarty, M. (under review) Measuring Spatial Ability: Reevaluating and Revising Measures of Mental Rotation.

Munns, M. E., Nelson, M., He, C., Topete, A., Rashidi, M., Chrastil, E. R., & Hegarty, M. (under review) Everyday Spatial Experience, Video Games, and Their Influence on Spatial Abilities.

Zhou, M., He, C. Topete, A., **Munns, M. E.,** & Hegarty, M. (under review). Navigation Strategies in an Outdoor Environment: Effects of Distal Landmarks, Spatial Knowledge, and Spatial Anxiety.

Conference Talks

Munns, M.E. & Hegarty M. . (2025, November 20th – November 23rd). *Towards a Taxonomy of Small-Scale Spatial Abilities*. [Talk]. Psychonomics Annual Meeting. Denver, CO, US.

Munns, M. E. & Hegarty, M. (2025, July 12th – July 13th). *A Shared Spatial Mental Representation System for Navigation and Reasoning*. [Talk]. Visualization in Science and Education - Gordon Research Seminar: Navigating the Multidisciplinary Milieu of Research and Practice, Bates College, Lewiston, ME, US.

Munns, M. E., Topete, A., Weisberg, S. M., Newcombe, N. S., & Hegarty, M. (2024, June 25-28). *Measuring Individual Differences in Navigation: Refining the Virtual SILCton task with Item Response Theory*. [Symposium Talk]. Spatial Cognition, Dublin, Ireland.

Munns, M. E. & Hegarty, M. (2024, June 25-28). *Evaluating Psychometric Properties of Spatial Ability Measures using Item Response Theory: Tests of Mental Rotation, Cross Sectioning, and Paper Folding*. [Symposium Talk]. Spatial Cognition, Dublin, Ireland.

Munns, M. E., & Hegarty, M. *The Influence of Everyday Hobbies and Video Games on Different Types of Spatial Abilities*. (2023, July 31 – August 2). [Invited Talk] Spatial Reasoning in STEM Education Workshop. University of Canberra, Canberra, ACT, Australia.

Munns, M. E., Tranquada-Torres, B., Chrastil, E., & Hegarty, M. (2022). *Large-Scale vs Small-Scale Spatial Abilities: Development of a Broad Spatial Activities Questionnaire*. [Talk]. Annual Meeting of the Cognitive Science Society, Toronto, ON, Canada.

13 Poster presentations and 2 symposia organized

Leadership, Outreach, & Service

Reviewer for: Cognition; Cognitive Research: Principles and Implications; Humanities and Social Sciences Communications; Journal of Cognitive Psychology; Proceedings of the Annual Meeting of the Cognitive Science Society; Spatial Cognition and Computation

Mentorship: 8 undergraduate research assistants, 6 Access Grads mentees

Co-President of Learning Unity for Virtual Reality Research

2022 – present

Planning Committee: Cognition, Perception, and Cognitive Neuroscience Weekly Seminar

2022 – 2024

Panelist: Center for Science and Engineering Program (CSEP) *Grad School A–Z Workshop*

2025

Software: Developed a shell program for PsychoPy and Pavlovio that allows users to create custom online testing batteries without requiring coding knowledge
